Anna Vakarova

Department of Economics, University of California, Berkeley

anna_vakarova@berkeley.edu · annavakarova.github.io

RESEARCH FIELDS

Microeconomic Theory, Fair Division, General Equilibrium

EDUCATION

University of California, Berkeley 2027 (expected)
Ph.D. in Economics
Dissertation: *[Essays in Fair Division]*
Committee: Federico Echenique, Chris Shannon, Quitze Valenzuela-Stookey, Yuichiro Kamada

Kyiv School of Economics 2020
M.A. in Economic Analysis

Taras Shevchenko National University of Kyiv 2018
B.A. in International Economic Relations

RESEARCH

1. *The power of Sperner's Lemma in Fair Division* (Job Market Paper)
   This paper develops a general approach to fair and efficient division that does not rely on equilibrium existence. First, I prove existence of an allocation that is envy-free and weak Pareto efficient. The result applies to both problems with divisible resources and to indivisible-resource problems after appropriate convexification. It applies to problems with infinite-dimensional allocation spaces. Second, I provide an algorithm to find an approximately envy-free and approximately weak Pareto efficient allocations in polynomial time for a fixed number of agents. The problem of fairness and efficiency of dynamic allocations arises in transportation, infrastructure, natural resource management, computing. Yet these problems cannot be tackled by present day fair division techniques. Third, I combine the infinite-dimensional existence result with the finite-dimensional algorithm to allocate fairly, efficiently, and dynamically.
2. *Multi-Period House Allocation as a Public Good Problem* (Working paper)
   with Federico Echenique and Albin Erlanson
   We prove existence of envy-free and Pareto efficient allocation in a multi-period indivisible goods assignment problem. Since Birkoff-von-Neumann fails in this environment, the allocation problem cannot be reduced to individual-level consumption problem where total demand cannot exceed supply. Therefore, we have to treat economy-wide allocations as a public good. We provide a market-based implementation by showing that such an allocation can be supported as a Lindahl equilibrium. We then look at a large economy and show that for large enough economies, the incentives to misrepresent agents' preferences vanish.
3. *Implementation in the Cake-Cutting Problem* (Working paper)
   I study the relationship between fractional cake allocations — partitions of unity — and lotteries over deterministic partitions. The project provides an implementation justification for working with fractional allocations: every partition of unity can be implemented by a randomization over feasible deterministic partitions. Moreover, I show that there exists an efficient and envy-free allocation that can be implemented by a lottery with finite support for a broad class of preferences-measures (in particular, atomic).
4. *Approximately Marginal Contribution Allocations* (Work in progress)
   I build on the established connection between Marginal Contribution (MC) allocations and Walrasian

equilibrium, which claims that in particular environments with quasilinear preferences, the two notions coincide. Does local convergence to quasilinear utility make agents' utility approximately equal to marginal contribution to unweighted social welfare? Not in general: a two-type logarithmic example has a limiting strictly positive gap despite small income effects. I provide a sufficient condition: equilibrium and unrestricted welfare-maximizing allocations remain in a common compact interior region on which the utility gradients converge to quasilinear utility gradients.

PRESENTATIONS

European Meeting of the Econometric Society
University College Dublin, Dublin — August 2026

2nd Berlin PhD Conference in Economics
Berlin School of Economics, Berlin — July 2026

North American Summer Meeting of the Econometric Society
Emory University, Atlanta — June 2026

29th Coalition Theory Network Conference
University of Padua, Padua — May 2026

Graduate Women in Economic Theory Conference (poster)
Yale University, New Haven — March 2026

European Winter Meeting of the Econometric Society
University of Cyprus, Nicosia — December 2025

Theory Lunch, Department of Decision Sciences
Bocconi University, Milan — December 2025

CLIMB Retreat
University of California, Berkeley — November 2025

TEACHING

Game Theory (Ph.D.) — Spring 2023-2025
Teaching Assistant

Mathematical Tools for Economists (Ph.D.) — Fall 2023-2026
Teaching Assistant

Intermediate microeconomics (undergrad) — 2021 - Fall 2022
Teaching Assistant

FELLOWSHIPS AND AWARDS

Doctoral Completion Fellowship, UC Berkeley — 2026-2027

Departmental Fellowship, UC Berkeley, Department of Economics — 2020-2025

FUIB Merit Fellowship, Kyiv School of Economics — 2018-2020

PROFESSIONAL SERVICE

Referee: *Theoretical Economics* — 2026

OTHER INFORMATION

Languages: English, Ukrainian, basic German

Citizenship: Ukraine

Age: 29 years old

REFERENCES

Federico Echenique
Professor
UC Berkeley
fede@econ.berkeley.edu

Chris Shannon
Professor
UC Berkeley
cshannon@econ.berkeley.edu

Quitze Valenzuela-Stookey
Assistant Professor
UC Berkeley
quitze@berkeley.edu